

```
# Task1
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

food = ["Rice", "Milk", "Egg"]
calories = [200, 150, 100]

df = pd.DataFrame({"Food": food, "Calories": calories})

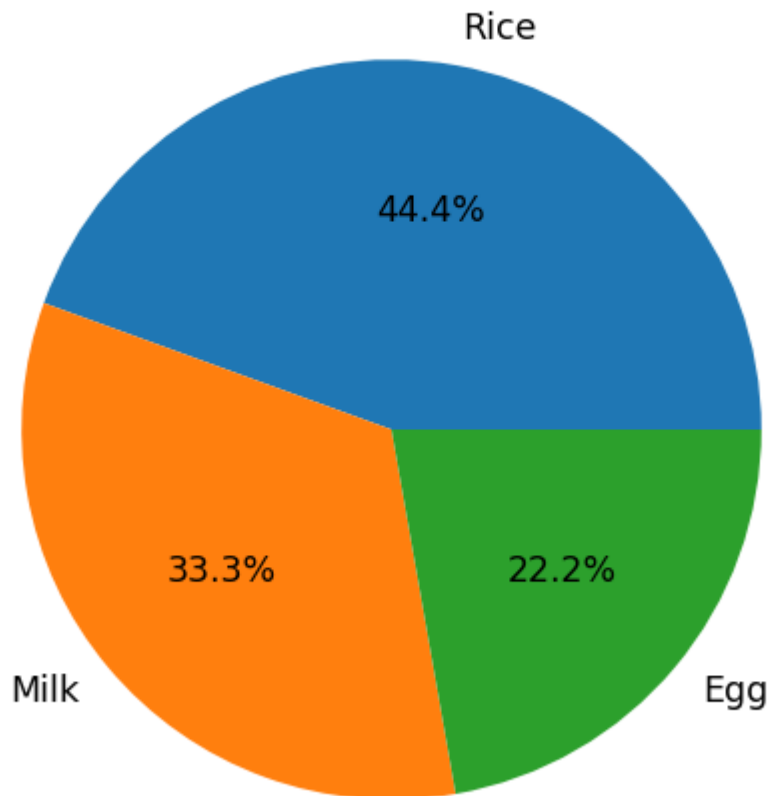
total = np.sum(df["Calories"])
print("Total Calories:", total)

plt.pie(df["Calories"], labels=df["Food"], autopct="%1.1f%%")
plt.title("Calorie Intake")
plt.show()

print("="*50)

===== RESTART: C:/Us
Total Calories: 450
```

Calorie Intake



```
# Task2
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

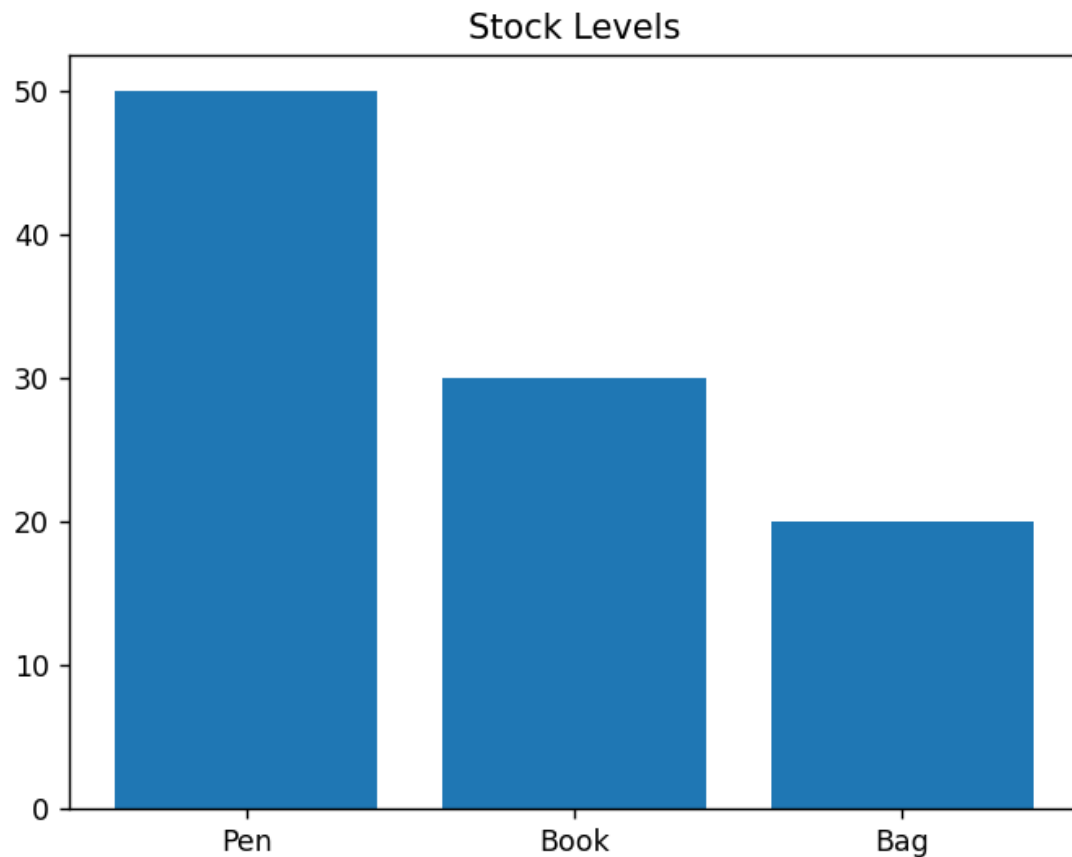
products = ["Pen", "Book", "Bag"]
quantity = [50, 30, 20]

df = pd.DataFrame({"Product": products, "Quantity": quantity})

total = np.sum(df["Quantity"])
print("Total Stock:", total)

plt.bar(df["Product"], df["Quantity"])
plt.title("Stock Levels")
plt.show()

print("="*50)
=====
Total Stock: 100
```



```
# Task3
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

players = ["A", "B", "C"]
scores = [80, 60, 90]

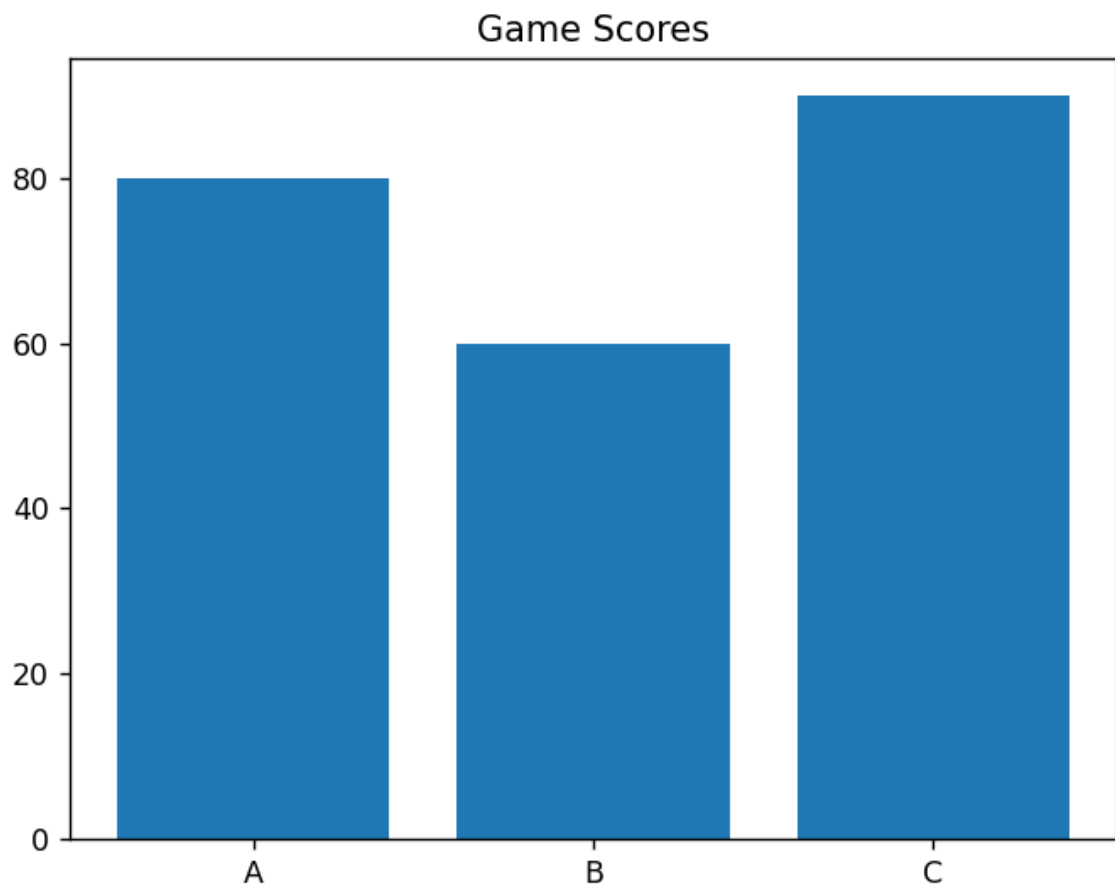
df = pd.DataFrame({"Player": players, "Score": scores})

print("Highest Score:", np.max(df["Score"]))

plt.bar(df["Player"], df["Score"])
plt.title("Game Scores")
plt.show()

print("="*50)

=====
Highest Score: 90
```



```
# Task4
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

hours = ["9AM", "10AM", "11AM", "12PM"]
vehicles = [50, 80, 120, 90]

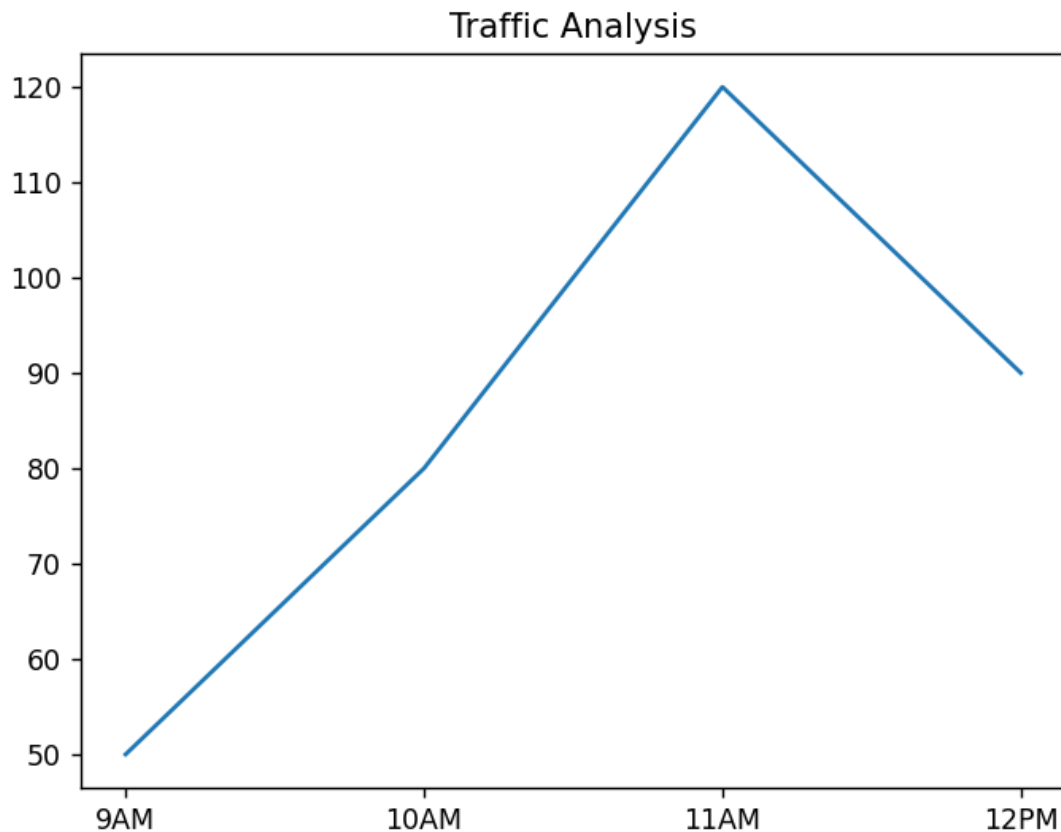
df = pd.DataFrame({"Time": hours, "Vehicles": vehicles})

peak = df.iloc[np.argmax(df["Vehicles"])]
print("Peak Time:", peak["Time"])

plt.plot(df["Time"], df["Vehicles"])
plt.title("Traffic Analysis")
plt.show()

print("="*50)

=====
Peak Time: 11AM
```



```
# Task5
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```
TP, FP, FN, TN = 50, 5, 10, 35
```

```
total = TP + FP + FN + TN
```

```
accuracy = (TP + TN) / total
```

```
precision = TP / (TP + FP)
```

```
recall = TP / (TP + FN)
```

```
f1 = 2 * (precision * recall) / (precision + recall)
```

```
print("Accuracy:", accuracy)
```

```
print("Precision:", precision)
```

```
print("Recall:", recall)
```

```
print("F1 Score:", f1)
```

```
print("="*50)
```

```
Accuracy: 0.85
```

```
Precision: 0.9090909090909091
```

```
Recall: 0.8333333333333334
```

```
F1 Score: 0.8695652173913043
```

```
# Task6
TP = [40,30,40]
FP = [10,5,10]
FN = [5,10,5]

for i in range(3):
    precision = TP[i]/(TP[i]+FP[i])
    recall = TP[i]/(TP[i]+FN[i])
    f1 = 2*(precision*recall)/(precision+recall)
    print(f"Class {i}: F1 =", f1)

print("="*50)
```

```
Class 0: F1 = 0.8421052631578948
Class 1: F1 = 0.7999999999999999
Class 2: F1 = 0.8421052631578948
```

```
# NUMPY TASKS
import numpy as np

# Task1
arr = np.arange(1,11)
print(arr)
print("="*50)

# Task2
data = np.array([10,20,30,40,50])
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Std Dev:", np.std(data))
print("="*50)

# Task3
rand_nums = np.random.rand(10)
print(rand_nums)
print("="*50)

# Task4
a = np.arange(1,7)
reshaped = a.reshape(2,3)
print(reshaped)
print("="*50)

# Task5
print("Max:", np.max(data))
print("Min:", np.min(data))
print("="*50)

# Task6
zeros = np.zeros((3,3))
ones = np.ones((3,3))
print(zeros)
print(ones)
print("="*50)
```

```

=====
[ 1  2  3  4  5  6  7  8  9 10]
=====
Mean: 30.0
Median: 30.0
Std Dev: 14.142135623730951
=====
[0.91924616 0.65987213 0.78248738 0.76214447 0.41730761 0.14608234
 0.71728479 0.77828626 0.99495907 0.86624929]
=====
[[1 2 3]
 [4 5 6]]
=====
Max: 50
Min: 10
=====
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]

```



```

# PANDAS TASKS
import pandas as pd

# Task1
df = pd.DataFrame({
    "Name": ["Asha", "Ravi", "John", "Meena"],
    "Marks": [85, 90, 78, 88]
})
print(df)
print("="*50)

# Task2
df["Grade"] = ["A", "A+", "B", "A"]
print(df)
print("="*50)

# Task3
sorted_df = df.sort_values(by="Marks")
print(sorted_df)
print("="*50)

# Task4
data = pd.read_csv("employee_data.csv")
print(data.head(25))
print("="*50)

# Task5
clean = data.dropna()
print(clean)
print("="*50)

# Task6
data.fillna(data.median(numeric_only=True), inplace=True)
print(data)
print("="*50)

```

	Name	Marks
0	Asha	85
1	Ravi	90
2	John	78
3	Meena	88

	Name	Marks	Grade
0	Asha	85	A
1	Ravi	90	A+
2	John	78	B
3	Meena	88	A

	Name	Marks	Grade
2	John	78	B
0	Asha	85	A
3	Meena	88	A
1	Ravi	90	A+

	Name	Age	Department
0	Alice	25	HR
1	Bob	30	Finance
2	Charlie	28	IT

	Name	Age	Department
0	Alice	25	HR
1	Bob	30	Finance
2	Charlie	28	IT

	Name	Age	Department
0	Alice	25	HR
1	Bob	30	Finance
2	Charlie	28	IT

```
# MATPLOTLIB TASKS
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr", "May"]
sales = [100, 150, 200, 180, 220]

# Task1
plt.plot(months, sales)
plt.title("Monthly Sales")
plt.show()
print("="*50)

# Task2
students = ["A", "B", "C", "D"]
marks = [80, 90, 70, 85]

plt.bar(students, marks)
plt.title("Student Marks")
plt.show()
print("="*50)

# Task3
height = [150, 160, 170, 180]
weight = [50, 60, 65, 75]

plt.scatter(height, weight)
plt.title("Height vs Weight")
plt.show()
print("="*50)

# Task4
ages = [18, 20, 22, 22, 23, 25, 27, 30]

plt.hist(ages)
plt.title("Age Distribution")
plt.show()
print("="*50)

# Task5
plt.plot(months, sales, label="Sales")
plt.legend()
plt.show()
print("="*50)
```

```
# Task6
sales2 = [90,120,160,170,200]

plt.plot(months,sales,label="2024")
plt.plot(months,sales2,label="2025")
plt.legend()
plt.show()
print("="*50)
|
```

